

Arman Javan Sekhavat Pishkhani

TEHRAN, IRAN

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EDUCATION

Master's Degree, Mechatronics Engineering

University of Tehran

2024–2027

TEHRAN, IRAN

- GPA: -
- Thesis Title: Simulation and Intelligent Control of an Endotracheal Intubation Robot
- Thesis Grade: -
- Supervisor: Dr. Khalil Alipour

Bachelor's Degree, Mechanical Engineering

K. N. Toosi University of Technology

2020–2024

TEHRAN, IRAN

- GPA: 3.79/4 (17.95/20) – 3.81/4 (17.99/20) Last two years
- Thesis Title: Design of a Controller for a Robotic Bicycle Using Intelligent Control Techniques
- Thesis Grade: A (19.00/20)
- Supervisor: Dr. S. Hossein Sadati

High School Diploma, STEM

National Organization for Development of Exceptional Talents (SAMPAD)

2017–2020

RASHT, GUILAN

RESEARCH INTERESTS

- Robotics
- Reinforcement Learning
- Control Theory

TEACHING EXPERIENCE

Teaching Assistant, Machine Vision

Department of Interdisciplinary Sciences and Technologies, University of Tehran

Presented by Dr. Hanieh Naderi

Sep. 2025 – Jan. 2026

- Designed problems for assignments and exams
- Graded the course assignments

TEACHING EXPERIENCE

Teaching Assistant, Automatic Control

Sep. 2023 – Jan. 2024

Mechanical Engineering Department, K. N. Toosi University of Technology

Presented by Dr. Naser Naserifar

- Conducted exercise-solving classes
- Produced educational materials

Teaching Assistant, Computer Programming

Feb. 2022 – Jun. 2022

Mechanical Engineering Department, K. N. Toosi University of Technology

Presented by Dr. Farschad Torabi

- Conducted exercise-solving classes focused on programming in C++ using the Qt framework
- Produced educational materials

TECHNICAL AND RESEARCH EXPERIENCE

- **Physics-informed Data-driven Modeling of a Two-DoF Robot Arm** [GitHub]
Utilized JAX for an optimized implementation of Lagrangian Neural Networks. This model was used for learning the dynamics of a two-DoF robot arm.
- **Simulation and Control of a 5RP Robot Arm** [GitHub]
Design, simulation, and implementation of various control techniques on a 5RP robotic manipulator.
- **Design of a Controller for a Robotic Bicycle Using Intelligent Control Techniques** [GitHub]
Designed linear and nonlinear controllers for a robotic bicycle using the DDPG algorithm. The robot was simulated in MuJoCo.
- **Mobile Robot Control System (teamwork)** [GitHub]
Designed a PID-based control system for a physical differential-drive mobile robot. A GUI application served as a control panel. The user was able to draw the desired path of the robot in the control panel.
- **Passive Steering Wheel (teamwork)** [GitHub]
Designed a CNN model capable of estimating the steering angle from an input image of a steering wheel.

PUBLICATIONS

- **Gray-Box Computed Torque Control for Differential-Drive Mobile Robot Tracking** [arXiv] [GitHub]
- **Optimal Intelligent Control of Robotic Manipulators with Optimal Reliance on Physics Information** [TechRxiv]
- **LiteMish: A Computationally Efficient and Smooth Algebraic Alternative to Mish** [TechRxiv][GitHub]

TECHNICAL SKILLS

- Programming languages and mathematical packages: C, C++, MATLAB, Python
- Software packages: Git, Proteus, LTSpice, STM32CubeMX, Keil uVision
- Software libraries and frameworks: Keras, TensorFlow, JAX, OpenCV, MuJoCo, Qt (GUI development)
- Computer aided design: SOLIDWORKS
- Development boards: STM32, ESP32, Arduino
- Operating systems: Windows, Linux
- Hardware skills: Soldering
- Other: Deep RL, Simulink, Network Programming, L^AT_EX

COURSES STUDIED

Course Name	Grade (out of 20)
Advanced Robotics	19.83
Machine Vision	20.00
Optimal Control	18.75
Modern Control	18.80
Fundamentals of Electrical Engineering I	20.00
Fundamentals of Electrical Engineering II	19.50
Neural Networks	19.00
Measurement and Control Systems	18.50
Engineering Drawing II	19.60
Introduction to Mechatronics	18.00

CURRENT RESEARCH

Developing a novel nonlinear control technique to enforce robust asymptotic stability of uncertain dynamical systems controlled by reinforcement learning agents.

HONORS AND AWARDS

- **Ranked within the top 10% of mechanical engineering students from the same year of entrance**
Issued by K. N. Toosi University of Technology • Sep. 2024
- **Ranked within the top 1.5% in the Iranian University Entrance Exam**
Issued by National Organization of Educational Testing • Sep. 2020

LANGUAGES

- **English**
Proficient
TOEFL iBT Score: 90
- **Persian**
Native language

REFERENCES

Khalil Alipour, Associate Professor

College of Interdisciplinary Science and Technology
University of Tehran
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Majid Sorouri, Assistant Professor

Department of Electronic Engineering
Maynooth University
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Farschad Torabi, Associate Professor

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